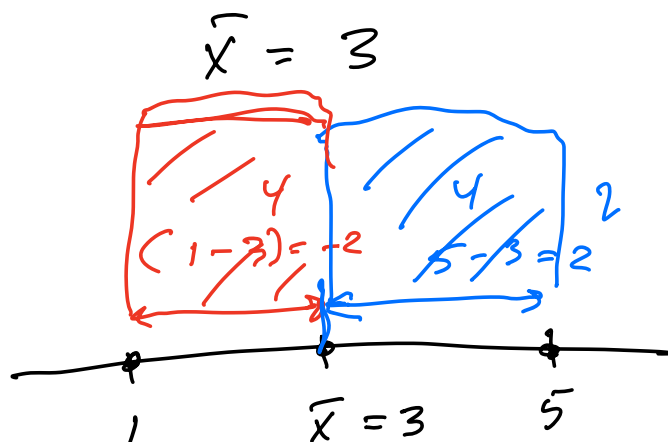


$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i = \frac{\sum x_i}{N}$$

$$\bar{x} = \frac{1}{N} x_1 + \frac{1}{N} x_2 + \dots + \frac{1}{N} x_N$$

$$x \in [1, 3, 5]$$



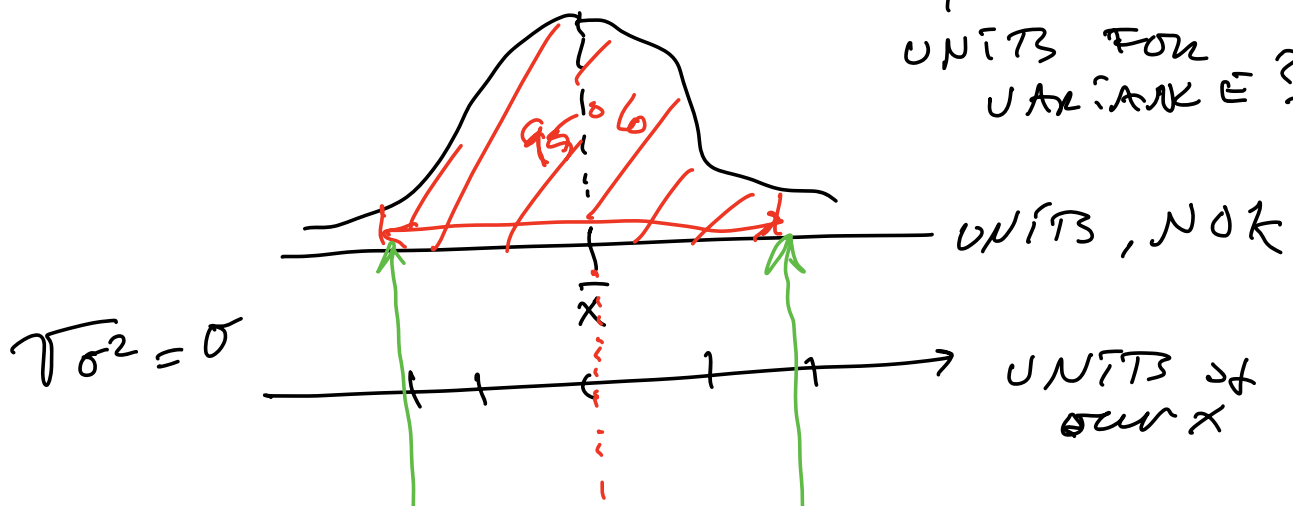
$$\sum (x_i - \bar{x}) = 0!$$

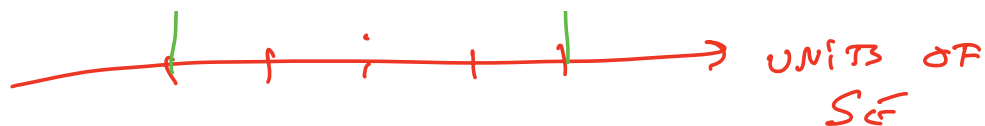
$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

Just a number $\rightarrow$

$$\sigma^2 = \frac{4 + 0 + 4}{N}$$

$\uparrow$
UNITS FOR VARIANCE?

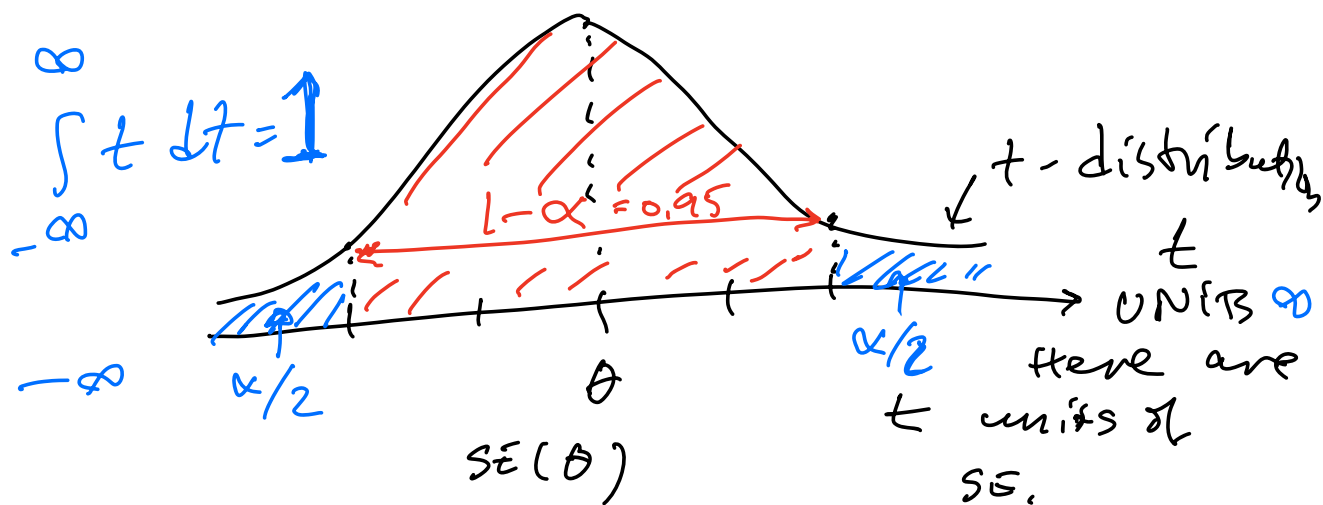




ANY CI is:
 AN ESTIMATE $\rightarrow \theta \pm CV \cdot SE(\theta)$
 CRITICAL VALUE $\rightarrow CV$
 STANDARD ERROR $\rightarrow SE(\theta)$

CV with a specific density distribution and often a specific degrees of freedom.

A CONFIDENCE INTERVAL: CI



$1-\alpha = 0.95$
 $\alpha = 0.05$
 $\uparrow$
 Level of significance